Chapitre 6 : La comparaison à une valeur théorique

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 6 sur ***la comparaison à une valeur théorique*** du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2ème édition).

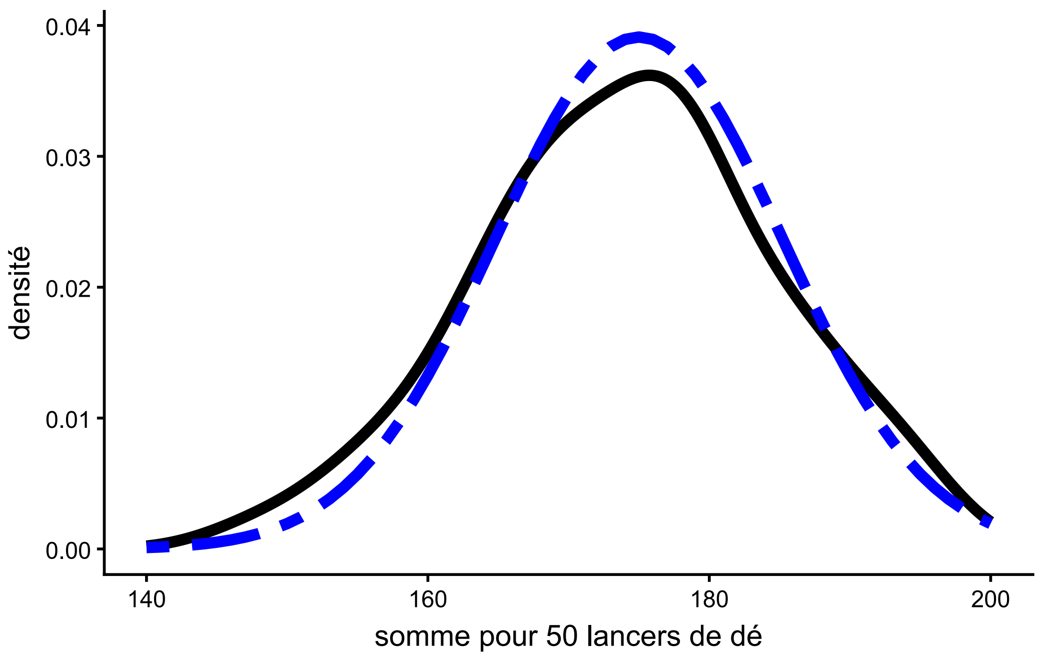
```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.0     ✔ readr     2.1.6
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.2     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ─────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

2 Théorème central limite

```
lancers_des <-
  purrr::map(
    .x = c(1:100),
    .f = ~ tibble(
        somme_des =
          runif(n = 50, min = 1, max = 6) |>
          round() |>
          sum()
      )
  ) |>
  bind_rows()

loi_normale <- tibble(
  x = seq(140, 200, 1),
  y = dnorm(x, mean = 175, sd = sqrt(104))
)

lancers_des |>
  ggplot() +
  aes(x = somme_des) +
```

```r
  geom_density(linewidth = 2) +
  geom_line(
    aes(x = x, y = y),
    data  = loi_normale,
    color = "blue",
    linetype = "twodash",
    linewidth = 2
    ) +
  labs(x = "somme pour 50 lancers de dé", y = "densité") +
  theme_classic()
```




```r
ggsave("img/06Rl01N.png", width = 14, height = 10, units = "cm")
```

3 Comparaison d'une proportion a une valeur théorique

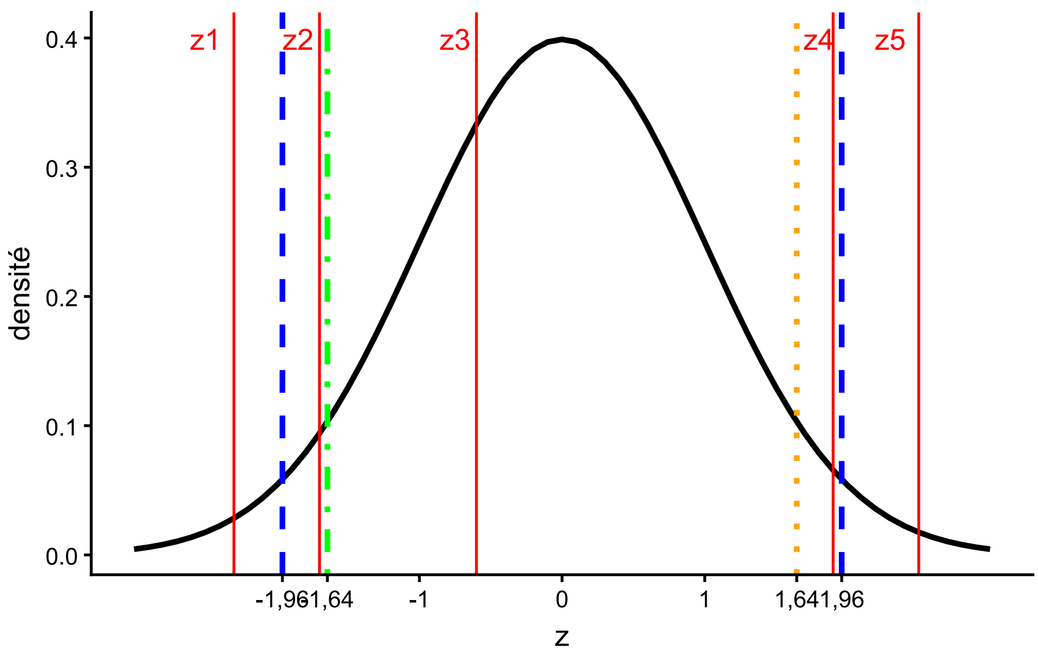
3.1 Exemple de comparaison

```r
tibble(
  z = seq(-3, 3, 0.1),
  "densité" = dnorm(z, mean = 0, sd = 1)
  ) |>
  ggplot() +
  aes(
```

```
    x = z,
    y = `densité`
  ) +
  geom_line(size = 1) +
  geom_vline(
    xintercept = qnorm(0.025),
    linewidth = 1,
    linetype = "dashed",
    color = "blue"
    ) +
  geom_vline(
    xintercept = qnorm(0.975),
    linewidth = 1,
    linetype = "dashed",
    color = "blue"
    ) +
  geom_vline(
    xintercept = qnorm(0.05),
    linewidth = 1,
    linetype = "dotdash",
    color = "green"
    ) +
  geom_vline(
    xintercept = qnorm(0.95),
    linewidth = 1,
    linetype = "dotted",
    color = "orange"
    ) +
  geom_vline(
    xintercept = -2.3,
    color = "red"
    ) +
  geom_vline(
    xintercept = -1.7,
    color = "red"
    ) +
  geom_vline(
    xintercept = -0.6,
    color = "red"
    ) +
  geom_vline(
    xintercept = 1.9,
    color = "red"
```

```
    ) +
  geom_vline(
    xintercept = 2.5,
    color = "red"
    ) +
  annotate("text", x = -2.5, y = 0.4, label = "z1", color = "red") +
  annotate("text", x = -1.85, y = 0.4, label = "z2", color = "red") +
  annotate("text", x = -0.75, y = 0.4, label = "z3", color = "red") +
  annotate("text", x = 1.8, y = 0.4, label = "z4", color = "red") +
  annotate("text", x = 2.3, y = 0.4, label = "z5", color = "red") +
  scale_x_continuous(
    label = c("-1,96", "-1,64","-1", "0", "1", "1,64", "1,96"),
    breaks = c(qnorm(0.025), qnorm(0.05), -1, 0, 1, qnorm(0.95), qnorm(0.975))
  ) +
  theme_classic()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```




```
ggsave("img/06Rl02N.png", width = 14, height = 10, units = "cm")
```

3.2 Exemple sur les pièces défectueuses

comparaison a une proportion test
Les valeurs :

```
nbr_piece_defecteuse <- 22
p_seuil <- 0.2
n <- 100
```

statistique de test calculée à la main

```
(z <- (nbr_piece_defecteuse/n - p_seuil)/sqrt((p_seuil*(1-p_seuil))/n))
```

```
[1] 0.5
```

valeur seuil

```
qnorm(0.95)
```

```
[1] 1.644854
```

utilisation de la fonction prop.test

```
prop.test(
  x = nbr_piece_defecteuse,
  n = n,
  p = p_seuil,
  alternative = "greater"
)
```

```

    1-sample proportions test with continuity correction

data:  nbr_piece_defecteuse out of n, null probability p_seuil
X-squared = 0.14062, df = 1, p-value = 0.3538
alternative hypothesis: true p is greater than 0.2
95 percent confidence interval:
 0.1554081 1.0000000
sample estimates:
   p
0.22
```

utilisation de la fonction `binom.test()`

```r
binom.test(
  x = nbr_piece_defecteuse,
  n = n,
  p = p_seuil,
  alternative = "greater"
)
```

```


	Exact binomial test

data:  nbr_piece_defecteuse and n
number of successes = 22, number of trials = 100, p-value = 0.346
alternative hypothesis: true probability of success is greater than 0.2
95 percent confidence interval:
 0.153889 1.000000
sample estimates:
probability of success 
                  0.22 
```

3.3 Exemple sur l'assemblée nationale

```r
library(rvest)
```

```


Attachement du package : 'rvest'
```

```
L'objet suivant est masqué depuis 'package:readr':

    guess_encoding
```

```r
html <- read_html("https://www2.assemblee-nationale.fr/deputes/liste/cat-sociopro/(vue)/tableau")

(prop_ass_nat <- html |>
  html_elements("table") |>
  html_table() |>
  bind_rows() |>
  count(Civilité))
```

```
# A tibble: 2 × 2
  Civilité     n
  <chr>    <int>
1 M.         367
2 Mme        207
```

calcul de la proportion d'hommes à l'Assemblée

```r
(p_ass <- prop_ass_nat[[1,2]] / sum(prop_ass_nat$n))
```

```
[1] 0.6393728
```

population française 1er janvier 2026 : https://www.insee.fr/fr/statistiques/2381474

```r
(p_pop <- 33551000 / (35531000 + 33551000))
```

```
[1] 0.4856692
```

calcul de la statistique de test

```r
(z <- (p_ass - p_pop)/sqrt((p_pop*(1-p_pop))/sum(prop_ass_nat$n)))
```

```
[1] 7.367981
```

comparaison aux valeurs de seuil

```r
z < qnorm(0.025)
```

```
[1] FALSE
```

```r
z > qnorm(0.975)
```

```
[1] TRUE
```

utilisation de la fonction `prop.test()`

```r
prop.test(
  x = p_ass,
  n = sum(prop_ass_nat$n),
```

```
  p = p_pop,
  alternative = "two.sided"
)
```

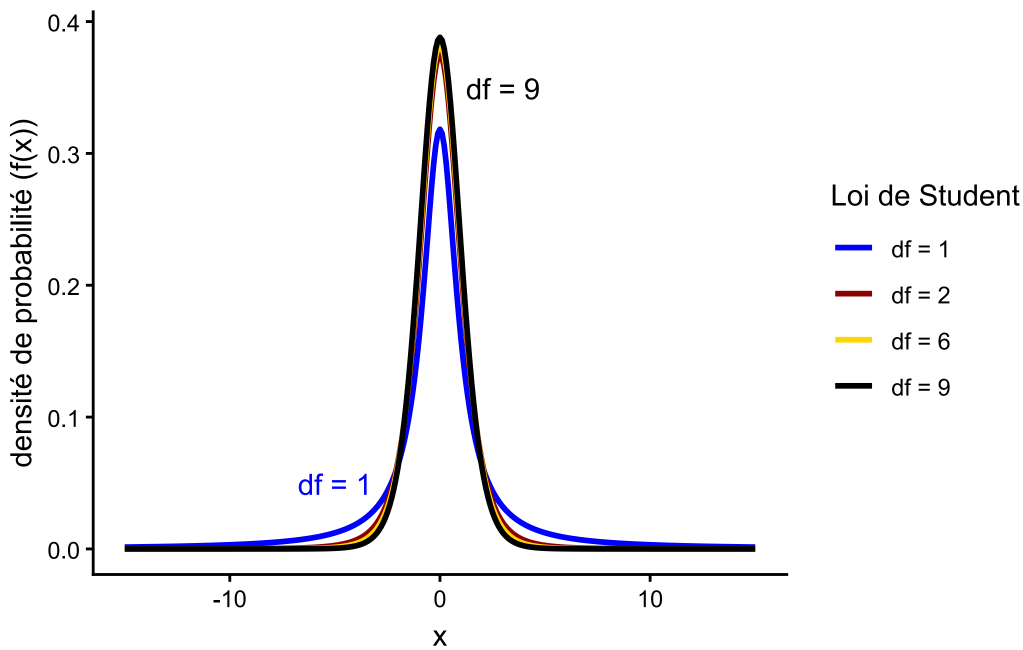
```

	1-sample proportions test with continuity correction

data:  p_ass out of sum(prop_ass_nat$n), null probability p_pop
X-squared = 537.59, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.4856692
95 percent confidence interval:
 8.222931e-06 1.020808e-02
sample estimates:
         p
0.00111389
```

4 Comparaison d'une moyenne à une valeur théorique

4.1 distribution de la loi de Student

```
# choix de l'intervalle de représentation
intervalle <- seq(-15, 15, 0.1)

# loi de Student de dégré de liberté égale à 1, 2, 6 et 9
tibble(
  loi_student =
    rep(
      c(
        "df = 1",
        "df = 2",
        "df = 6",
        "df = 9"
      ),
      each = length(intervalle)
    ),
  x = rep(intervalle, 4),
  densite =
    c(
      dt(intervalle, df = 1),
      dt(intervalle, df = 4),
      dt(intervalle, df = 6),
      dt(intervalle, df = 9)
```

```
    )
) |>
  ggplot() +
  aes(x = x, y = densite, color = loi_student) +
  geom_line(linewidth = 1) +
  annotate("text", x = -5, y = 0.05, label = "df = 1", color = "blue") +
  annotate("text", x = 3, y = 0.35, label = "df = 9", color = "black") +
  theme_classic() +
  scale_color_manual(values = c("blue", "dark red", "gold", "black")) +
  ylab("densité de probabilité (f(x))") +
  labs(color = "Loi de Student")
```




```
ggsave("img/06Rl06N.png", width = 14, height = 10, units = "cm")
```

4.2 Exemple d'utilisation du test de Student

calcul_moyenne

```
mean(women$weight)
```

```
[1] 136.7333
```

vérification de la normalité de la variable

```
shapiro.test(women$weight)
```

```

	Shapiro-Wilk normality test

data:  women$weight
W = 0.96036, p-value = 0.6986
```

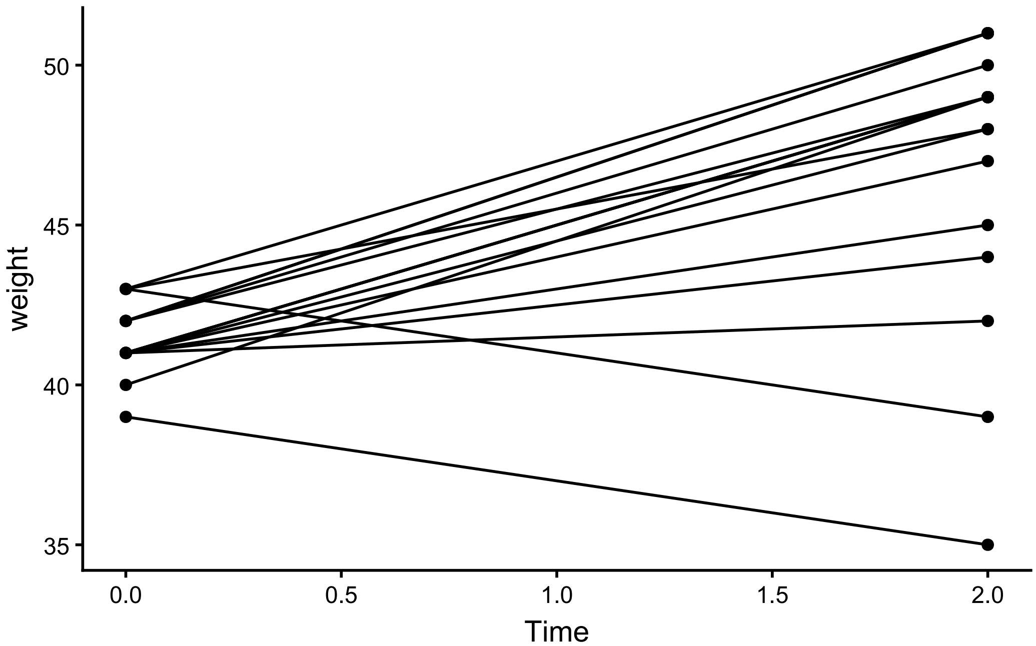
test de Student, la masse moyenne de l'échantillon est-elle égale à celle de la population ?

```
t.test(
  x = women$weight,
  mu = 79.3*2.2
)
```

```

	One Sample t-test

data:  women$weight
t = -9.4276, df = 14, p-value = 1.929e-07
alternative hypothesis: true mean is not equal to 174.46
95 percent confidence interval:
 128.1504 145.3162
sample estimates:
mean of x
 136.7333
```

4.3 Test de Student apparié

```
ChickWeight |>
  filter(
    Diet == 1,
    Time %in% c(0, 2)
    ) |>
  ggplot() +
  aes(x = Time, y = weight, group = Chick) +
  geom_point() +
  geom_line() +
  theme_classic()
```



création données

```
(poulet_croissance <- ChickWeight |>
  filter(
    Diet == 1,
    Time %in% c(0, 2)
    ) |>
  pivot_wider(
    id_cols = !Diet,
    names_from = Time,
    values_from = weight
  ) |>
  mutate(difference = `2` - `0`))
```

```
# A tibble: 20 × 4
   Chick   `0`   `2` difference
   <ord> <dbl> <dbl>      <dbl>
 1 1        42    51          9
 2 2        40    49          9
 3 3        43    39         -4
 4 4        42    49          7
 5 5        41    42          1
 6 6        41    49          8
 7 7        41    49          8
```

```
 8 8        42    50          8
 9 9        42    51          9
10 10       41    44          3
11 11       43    51          8
12 12       41    49          8
13 13       41    48          7
14 14       41    49          8
15 15       41    49          8
16 16       41    45          4
17 17       42    51          9
18 18       39    35         -4
19 19       43    48          5
20 20       41    47          6
```

```
# test de normalité des données
shapiro.test(poulet_croissance$difference)
```

```

	Shapiro-Wilk normality test

data:  poulet_croissance$difference
W = 0.7411, p-value = 0.0001293
```

```
# test de Student apparié à ne pas faire normalement
t.test(poulet_croissance$`0`, poulet_croissance$`2`, paired = TRUE)
```

```

	Paired t-test

data:  poulet_croissance$`0` and poulet_croissance$`2`
t = -6.5346, df = 19, p-value = 2.94e-06
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -7.72375 -3.97625
sample estimates:
mean difference 
          -5.85 
```

4.4 Test de Wilcoxon apparié

```r
# ajout des rangs signés
(test_wilcoxon <- poulet_croissance |>
  filter(difference != 0) |>
  arrange(abs(difference)) |>
  rowid_to_column(var = "rang") |>
  mutate(rang_signe = rang*sign(difference)))
```

```
# A tibble: 20 × 6
    rang Chick   `0`   `2` difference rang_signe
   <int> <ord> <dbl> <dbl>      <dbl>      <dbl>
 1     1 5        41    42          1          1
 2     2 10       41    44          3          2
 3     3 3        43    39         -4         -3
 4     4 16       41    45          4          4
 5     5 18       39    35         -4         -5
 6     6 19       43    48          5          6
 7     7 20       41    47          6          7
 8     8 4        42    49          7          8
 9     9 13       41    48          7          9
10    10 6        41    49          8         10
11    11 7        41    49          8         11
12    12 8        42    50          8         12
13    13 11       43    51          8         13
14    14 12       41    49          8         14
15    15 14       41    49          8         15
16    16 15       41    49          8         16
17    17 1        42    51          9         17
18    18 2        40    49          9         18
19    19 9        42    51          9         19
20    20 17       42    51          9         20
```

```r
# calcul des paramètres de la statistique de test
(n <- nrow(test_wilcoxon))
```

```
[1] 20
```

```r
(somme_rang_positif <- test_wilcoxon |>
  filter(rang_signe > 0) |>
  summarise(sum(rang_signe)))
```

```
# A tibble: 1 × 1
  `sum(rang_signe)`
              <dbl>
1               202
```

```
(somme_rang_negatif <- test_wilcoxon |>
  filter(rang_signe < 0) |>
  summarise(sum(rang_signe)))
```

```
# A tibble: 1 × 1
  `sum(rang_signe)`
              <dbl>
1                -8
```

```
(T_calcule <- min(abs(somme_rang_positif), abs(somme_rang_negatif)))
```

```
[1] 8
```

```
# calcul de la statistique de test
(z <- (T_calcule-(n*(n+1))/4)/sqrt((n*(n+1)*(2*n+1))/24))
```

```
[1] -3.621269
```

```
# conclusion
z < qnorm(0.05)
```

```
[1] TRUE
```

```
# test de wilcoxon apparié
wilcox.test(poulet_croissance$`0`, poulet_croissance$`2`, paired = TRUE)
```

```
Warning in wilcox.test.default(poulet_croissance$`0`, poulet_croissance$`2`, :
impossible de calculer la p-value exacte avec des ex-aequos
```

```


    Wilcoxon signed rank test with continuity correction

```

```
data:  poulet_croissance$`0` and poulet_croissance$`2`
V = 8, p-value = 0.0002889
alternative hypothesis: true location shift is not equal to 0
```

5 Variance comparer à une valeur théorique

calcul de la variance

```
var(women$weight)
```

```
[1] 240.2095
```

comparaison d'une variance à une valeur de reférence

```
# données
n <- nrow(PlantGrowth)
var_echantillon <- var(PlantGrowth$weight)
var_theo <- 0.35

# calcul de la statisique de test
((n-1)*var_echantillon/var_theo)
```

```
[1] 40.73837
```

```
# récupération de la valeur seuil pour alpha = 0,05
qchisq(p = 1-0.05, df = n-1)
```

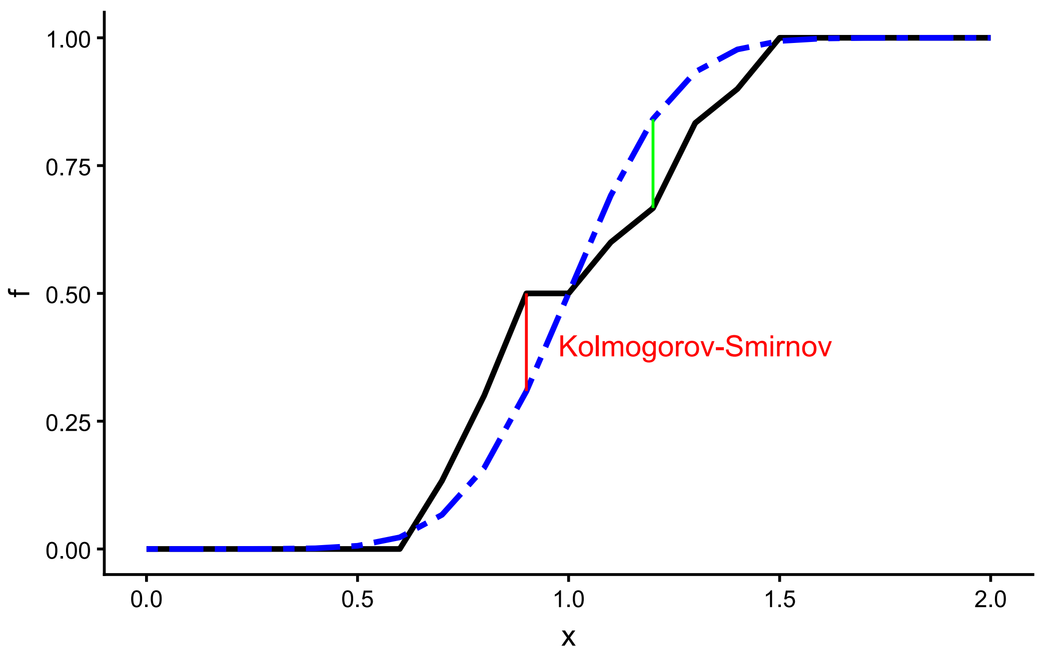
```
[1] 42.55697
```

6 Test de Kolmogorov Smirnov

comparaison avec le test de kuiper

```
set.seed(1903)

echantillon <- 0.5 + 1.1 * runif(30)
fonction_repartition <- ecdf(echantillon)

tibble(
  x = seq(0, 2, 0.1),
  f = fonction_repartition(x),
```

```
  densite_normale = pnorm(x, mean = 1, sd = 0.2)
  ) |>
  ggplot() +
  aes(x = x, y = f) +
  geom_line(linewidth = 1) +
  geom_line(
    aes( y = densite_normale),
    color = "blue",
    linewidth = 1,
    linetype = "twodash"
  ) +
   annotate("segment", x = 1.2, xend = 1.2, y = 0.667, yend = 0.841, color =
"green") +
  annotate("segment", x = 0.9, xend = 0.9, y = 0.309, yend = 0.5, color = "red") +
  annotate("text", x = 1.3, y = 0.4, color = "red", label = "Kolmogorov-Smirnov") +
  theme_classic()
```



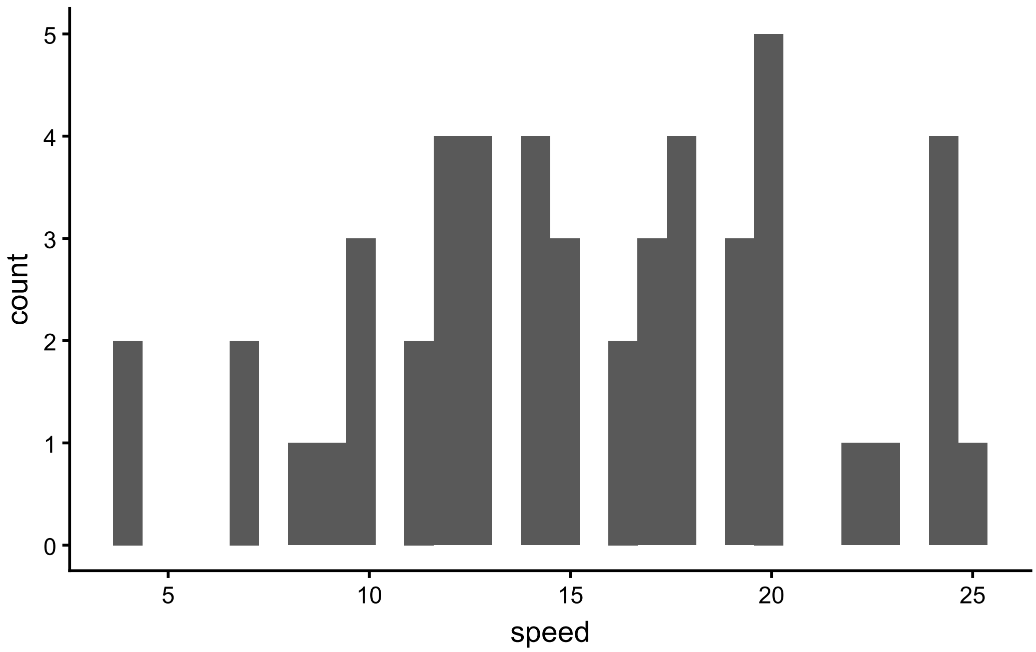

```
ggsave("img/06Rl18N.png", width = 14, height = 10, units = "cm")
```

test kolmogorov-smirnov

```
ggplot(cars) +
  aes(x = speed) +
```

```r
  geom_histogram() +
  theme_classic()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
ks.test(cars$speed, "pgamma", shape = 14)
```

```
Warning in ks.test.default(cars$speed, "pgamma", shape = 14): ties should not
be present for the one-sample Kolmogorov-Smirnov test
```

```

	Asymptotic one-sample Kolmogorov-Smirnov test

data:  cars$speed
D = 0.23913, p-value = 0.006572
alternative hypothesis: two-sided
```

```r
ks.test(cars$speed, "pgamma", shape = 15)
```

```
Warning in ks.test.default(cars$speed, "pgamma", shape = 15): ties should not
be present for the one-sample Kolmogorov-Smirnov test
```

```

    Asymptotic one-sample Kolmogorov-Smirnov test

data:  cars$speed
D = 0.17192, p-value = 0.1041
alternative hypothesis: two-sided
```

```
ks.test(cars$speed, "pnorm", mean = 15, sd = 2)
```

```
Warning in ks.test.default(cars$speed, "pnorm", mean = 15, sd = 2): ties should
not be present for the one-sample Kolmogorov-Smirnov test
```

```

    Asymptotic one-sample Kolmogorov-Smirnov test

data:  cars$speed
D = 0.31319, p-value = 0.0001099
alternative hypothesis: two-sided
```

```
ks.test(cars$speed, "pnorm", mean = 15, sd = 4)
```

```
Warning in ks.test.default(cars$speed, "pnorm", mean = 15, sd = 4): ties should
not be present for the one-sample Kolmogorov-Smirnov test
```

```

    Asymptotic one-sample Kolmogorov-Smirnov test

data:  cars$speed
D = 0.15337, p-value = 0.1901
alternative hypothesis: two-sided
```

test kolmogorov-smirnov comparaison

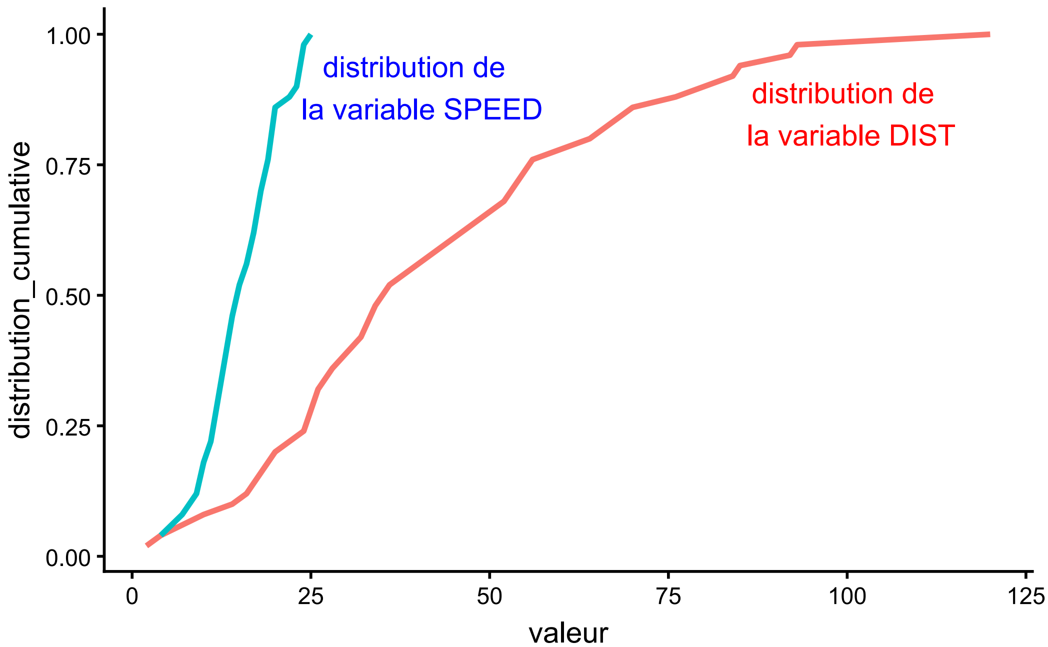
```
ks.test(cars$speed, cars$dist)
```

```

    Exact two-sample Kolmogorov-Smirnov test

data:  cars$speed and cars$dist
D = 0.76, p-value = 5.995e-15
alternative hypothesis: two-sided
```

graphique

```
fonction_repartition_speed <-  ecdf(cars$speed)
fonction_repartition_dist <-  ecdf(cars$dist)

bind_rows(
  cars |>
    mutate(
      distribution_speed = fonction_repartition_speed(speed),
    ) |>
    select(-dist) |>
    pivot_longer(
      !speed,
      names_to = "type_distribution",
      values_to = "distribution_cumulative"
    ) |>
    rename("valeur" = "speed"),
  cars |>
    mutate(
      distribution_dist = fonction_repartition_dist(dist),
    ) |>
    select(-speed) |>
    pivot_longer(
      !dist,
      names_to = "type_distribution",
      values_to = "distribution_cumulative"
    ) |>
    rename("valeur" = "dist")
) |>
  ggplot() +
  aes(x = valeur, y = distribution_cumulative, color = type_distribution) +
  geom_line(linewidth = 1) +
   annotate("text", x = 100, y = 0.85, color = "red", label = "distribution de
\n la variable DIST") +
  annotate("text", x = 40, y = 0.9, color = "blue", label = "distribution de \n
la variable SPEED") +
```

```
  theme_classic() +
  theme(legend.position = "none")
```




```
ggsave("img/06Rl25N.png", width= 14, height = 10, units = "cm")
```